

Mankiewicz Extension Theorem

MATHLIBANNEX / SELECTED READING SCOPE

The mathematical goal

An isometry equivalence from a nonempty open connected subset onto an open subset of real normed spaces extends uniquely to an affine isometry equivalence of the ambient spaces. The selected route also reaches convex-domain and ball specializations.

11 approved Cards in this selected reading scope. The catalog contains 188 distinct Cards across the four Projects.

Scope

Local distance information becomes a rigid affine structure. This Project exposes the reusable chain from reflection and midpoint geometry to global extension.

Hypotheses

Real normed spaces and an isometry equivalence. The principal theorem requires a nonempty open connected source and an open target. The convex specialization requires a convex source set with nonempty ambient interior and exact preservation of ambient-interior membership; the target set is not separately assumed convex. Exact assumptions, including radius conditions for balls, remain in the linked exact declarations.

Scope limits

The eleven declaration explanations are reviewed for source-exposition correspondence. This does not claim a line-by-line human proof audit. Boundary Inputs include Lean Core, Mathlib, and omitted MathlibAnnex support declarations.

Mathematical routes

Reflection symmetry identifies midpoints, and midpoint preservation yields local affine-isometry charts. Agreement on open overlaps glues these charts over the nonempty open connected domain. Convex and ball consequences keep their separate interior and radius hypotheses.

Each route shows its direct Cards and the reused prerequisites needed in the selected reading graph. Shared Cards count once within each route and once in the Catalog.

Reflection and midpoint rigidity

A bounded symmetric lens and reflection-center rigidity identify midpoints under an isometry. Containment in matching ambient balls supplies the local hypotheses for midpoint preservation.

5 direct Cards + 0 reused prerequisites = 5 unique Cards. Shared Cards count once in this route.

- Key result: [Midpoint preservation on a quarter ball](#)
- Key definition: [Symmetric lens](#)

[Read this route with prerequisites](#)

Local affine-isometry charts

Midpoint preservation yields preservation of affine segments on a smaller ball. The exact source path through omitted radial-map helpers constructs one ambient affine isometry agreeing with the given map there.

2 direct Cards + 5 reused prerequisites = 7 unique Cards. Shared Cards count once in this route.

- Key result: [Local affine-isometry chart on a smaller ball](#)
- Key result: [Affine-segment preservation on a quarter ball](#)

[Read this route with prerequisites](#)

Global extension and specializations

Local affine-isometry charts agree on open overlaps and glue over a nonempty open connected source. Continuity and interior matching yield the convex, open-ball and equal-radius closed-ball specializations with their stated hypotheses.

4 direct Cards + 7 reused prerequisites = 11 unique Cards. Shared Cards count once in this route.

- Key result: [Mankiewicz extension on open connected domains](#)
- Related specialization: [Affine extension from equal-radius closed balls](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#)

Level 0 (2 Cards)

Level 0

Symmetric lens

Names the intersection of two closed balls with a common radius.

`MathlibAnnex.symmetricLens`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Midpoint preservation on a symmetric lens](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Center transport under bounded point-reflection symmetry

Uses a doubling argument to identify the reflection centers under a bijective isometry.

`MathlibAnnex.IsometryEquiv.map_center_of_mapsTo_pointReflection`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Midpoint preservation on a symmetric lens](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (1 Card)

Level 1

Midpoint preservation on a symmetric lens

Identifies the centers of two lenses without prescribing images of their foci.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens`

Immediate Card prerequisites: [Center transport under bounded point-reflection symmetry](#) · [Symmetric lens](#)

Used by in this scope: [Midpoint preservation under lens containment](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2 (1 Card)

Level 2

Midpoint preservation under lens containment

Restricts a set isometry and its inverse to the two contained lenses.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_symmetricLens_subset`

Immediate Card prerequisites: [Midpoint preservation on a symmetric lens](#)

Used by in this scope: [Midpoint preservation on a quarter ball](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (1 Card)

Level 3

Midpoint preservation on a quarter ball

Supplies both lens inclusions from matching ambient balls.

`MathlibAnnex.IsometryEquiv.map_midpoint_of_mem_ball`

Immediate Card prerequisites: [Midpoint preservation under lens containment](#)

Used by in this scope: [Affine-segment preservation on a quarter ball](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4 (1 Card)

Level 4

Affine-segment preservation on a quarter ball

Passes from midpoint preservation to every real parameter in a segment.

`MathlibAnnex.IsometryEquiv.map_lineMap_of_mem_ball`

Immediate Card prerequisites: [Midpoint preservation on a quarter ball](#)

Used by in this scope: [Local affine-isometry chart on a smaller ball](#)

[Open Declaration Card](#) [Read exact source](#)

Level 5 (1 Card)

Level 5

Local affine-isometry chart on a smaller ball

Constructs one ambient surjective affine isometry from the same radial map throughout.

`MathlibAnnex.IsometryEquiv.exists_affineExtension_eqOn_ball`

Immediate Card prerequisites: [Affine-segment preservation on a quarter ball](#)

Used by in this scope: [Mankiewicz extension on open connected domains](#)

[Open Declaration Card](#) [Read exact source](#)

Level 6 (1 Card)

Level 6

Mankiewicz extension on open connected domains

Glues local affine-isometry charts using their agreement on open overlaps.

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension`

Immediate Card prerequisites: [Local affine-isometry chart on a smaller ball](#)

Used by in this scope: [Convex-set extension via ambient interiors](#) · [Affine extension from open balls](#)

[Open Declaration Card](#) [Read exact source](#)

Level 7 (2 Cards)

Level 7

Affine extension from open balls

Applies the open connected theorem to two positive-radius balls.

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_ball`

Immediate Card prerequisites: [Mankiewicz extension on open connected domains](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 7

Convex-set extension via ambient interiors

Extends agreement from a convex interior to all source points by continuity.

`MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_of_convex`

Immediate Card prerequisites: [Mankiewicz extension on open connected domains](#)

Used by in this scope: [Affine extension from equal-radius closed balls](#)

[Open Declaration Card](#) [Read exact source](#)

Level 8 (1 Card)

Level 8

Affine extension from equal-radius closed balls

Recovers the centers, matches the interiors, and then extends across the boundary.

```
MathlibAnnex.IsometryEquiv.existsUnique_affineExtension_closedBall
```

Immediate Card prerequisites: [Convex-set extension via ambient interiors](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Source exploration and boundary inputs

Full source declarations, helper paths and provider records are preserved separately from this selected Card reading scope.

[Explore 11 full source declarations](#) · [Earlier full-source Project PDF](#) · [Current selected reading PDF](#)